

Cooling Tower Optimisation

Task 6 (Group Report)



**Pierre BEDRUNE, Martin HARRAMBAT, Clémence-Philomène HINOT,
John HOARAU**

Prepared for March 6, 2026

Abstract

TO DO

Acronyms

BFGS Broyden-Fletcher-Goldfarb-Shanno

PSO Particle Swarm Optimisation

SA Simulated Annealing

Contents

1	Introduction	5
2	Cooling Tower	6
3	Algorithms	7
3.1	Particle Swarm Optimisation method	7
3.2	Simulated Annealing method	7
3.3	Broyden-Fletcher-Goldfarb-Shanno method	7
4	Results	8
4.1	Particle Swarm Optimisation results	8
4.2	Simulated Annealing results	8
4.3	Broyden-Fletcher-Goldfarb-Shanno results	8
4.4	Comparison between the three methods	8
4.5	Comparison of the methods performances	8
5	Discussion	9
6	Conclusion	10
7	Appendices	11
	Appendix A Code Listings	11
	Appendix A.A PSO implementation	11
	Appendix A.B SA implementation	11
	Appendix A.C BFGS implementation	11
	Appendix B Detailed Results Data	11

Appendix B.A PSO results	11
Appendix B.B SA results	11
Appendix B.C BFGS results	11

1 Introduction

TO DO

2 Cooling Tower

TO DO → formalising the problem

Objective: Minimising the surface of the cooling surface

Problem information:

- Design variables:
 - fixed volume
 - r_0 to r_m
- Constraint:
 - hyperboloid
- 8 simulations:
 - 1 imposed
 - x simulations with radii fixed and heights which vary
 - x simulations with height fixed and radii which vary
 - x simulations both heights and radii vary
- run 3 algorithms on each simulation: PSO, SA, and BFGS

Other information:

- hyperbolic regression with 3 points: r_0 , $r_{m/2}$, r_m
- new constraint: $r_m i$ need to be greater than $r_{m/2}$ and to be lower than r_m
- usually $r_0 < r_m$

- r_0 and r_m fixed arbitrary
- calculate the volume at each step of each algorithm
- to ensure the respect of the fixed volume we will use a Penalty function
- continuous problem

3 Algorithms

3.1 Particle Swarm Optimisation method

TO DO

3.2 Simulated Annealing method

TO DO

3.3 Broyden-Fletcher-Goldfarb-Shanno method

TO DO

4 Results

4.1 Particle Swarm Optimisation results

TO DO

4.2 Simulated Annealing results

TO DO

4.3 Broyden-Fletcher-Goldfarb-Shanno results

TO DO

4.4 Comparison between the three methods

TO DO

4.5 Comparison of the methods performances

TO DO

Explanations: As with Tasks 3 and 4, one aim of this Task is to compare the performance of the numerical techniques that you have employed by, for example, counting the number of function evaluations that are involved, recording the elapsed time, etc.

5 Discussion

TO DO → Results' discussion + Reflection on the work done

6 Conclusion

TO DO

7 Appendices

Appendix A Code Listings

Appendix A.A PSO implementation

TO DO

Appendix A.B SA implementation

TO DO

Appendix A.C BFGS implementation

TO DO

Appendix B Detailed Results Data

Appendix B.A PSO results

TO DO

Appendix B.B SA results

TO DO

Appendix B.C BFGS results

TO DO